Htrack26

Learning & Project Work

Technologies, tools, and skills explored while developing a Flutter-based health-data application.



Dart • Flutter • VS Code • Git • GitHub

1. Project Overview

What Htrack26 is designed to do

Htrack26 is a Flutter-based health-data tracking app. Its planned workflow helps users organize medical reports, review extracted values, visualize changes over time, and receive possible-trend alerts.

Report handling

Upload a report and review extracted test values.

Health tracking

Display historical results and identify possible trends.

Patient control

Plan controlled, time-limited sharing with doctors.

These are project goals from the report; not all features are claimed to be implemented.

2. Dart Programming Language

The language used to write Flutter application code

Variables & data

Represent values such as test names, dates, and numbers.

Functions

Group actions into reusable steps, such as validating a form.

Classes & objects

Organize related data and behavior.

How Dart relates to Htrack26

Dart is the programming language used by Flutter.

Dart code describes app behavior and user interactions.

Core concepts to practice: variables, functions, lists, classes, and async/await.

3. Flutter Framework

Building the mobile app interface

Widgets

Reusable building blocks for text, buttons, images, and screens.

Layouts

Arrange elements using rows, columns, padding, and containers.

Navigation

Move between Home, Upload, and Track Health screens.

Project application

Use Flutter to create Htrack26's patient-facing screens.

Forms can collect report details and let users confirm extracted values.

fl_chart is planned for historical health graphs.

Riverpod and go_router are planned for state management and navigation.

4. VS Code Development Environment

Writing, running, and debugging code

Editor

Create and edit Dart files, such as lib/main.dart.

Terminal

Run commands such as flutter pub get, flutter run, and git status.

Debugging

Read build output and use errors to locate problems.

Tools explored in the workflow

Use the integrated terminal to run Flutter and Git commands.

Check available devices and run the app on an emulator.

Use project folders to locate app code and Android configuration.

5. Git: Version Control

Tracking changes to project files

Git records changes over time so developers can review work and return to earlier versions when needed.

git status

Check which files are changed or untracked.

git add / git commit

Select changes and save a snapshot with a message.

git branch / checkout

Work with branches and switch between versions.

Git workflow skills

Inspect repository status and understand changed files.

Use commits as checkpoints for project progress.

Check the current branch and status before branch operations.

6. GitHub: Remote Repository & Collaboration

Hosting a Git repository online

Remote repository

An online copy of the project repository.

git clone / git pull

Get a repository or download newer remote changes.

git push

Upload local commits to the remote repository.

Git vs GitHub

Git is the version-control tool on your computer.

GitHub is an online platform that hosts Git repositories.

Typical workflow: edit → git status → git add → git commit → git push.

Only claim clone/push/pull as completed work if you actually used them on this repository.

7. Progress: Practiced vs Planned

Be accurate about what has been practiced and what remains

Explored / practiced

- Flutter setup and project execution
- Dart/Flutter project structure
- VS Code terminal and commands
- Git commands and branch concepts
- GitHub workflow concepts

Planned / to implement

- Htrack26 report upload and OCR
- Supabase database and authentication
- FastAPI analysis service
- Reviewed trend-detection rules
- FCM notifications and secure sharing

Before presenting, move items to “completed” only if you can demonstrate or explain your own work.

The project report describes the intended architecture; it does not prove each feature is already implemented.

8. Skills Learned & Next Milestone

Technical skills connected to the project

Understand how Dart and Flutter work together to build a mobile app.

Navigate project files and run/debug an app using VS Code.

Understand Git commits, branches, and repository status.

Understand how GitHub hosts a remote repository and supports collaboration.

Identify next steps to connect OCR, storage, analysis, and notifications.

Next milestone: demonstrate a simple Htrack26 screen, then implement report upload and value review using sample data.

Thank You

Htrack26 — Learning, implementation, and next steps

Questions & Discussion